

Technical Document #102: Database Systems - Comprehensive Overview

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Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Database caching stores frequently accessed data in memory.
- Relational databases organize data into tables with rows and columns.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.